6. Industrial Facilities, Reconstructions & Future Project Pipeline

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Primary OWL Classes: `IndustrialProject`, `Reconstruction`, `FutureProject`, `MidCenturyProject`, `ModernProject`, `ProjectPeriod`
Primary OWL Properties: `usesMaterial`, `usesMachine`, `hasSiteManager`, `year`

6.1 Industrial Projects (Class: IndustrialProject)

In the ontology, `IndustrialProject` is a direct subclass of `BuildingProject` alongside `ResidentialProject` and `CommercialProject`. It models manufacturing plants, automotive assembly halls, automated logistics warehouses, and energy infrastructure. Its formal OWL axioms are:

- `rdfs:subClassOf BuildingProject` (inheriting `hasCustomer some Customer` and `hasSiteManager some SiteManager`).
- `year some integer[≥ 2000, < 2026]` (standard completed industrial catalog projects fall in the `ModernProject` era).
- `usesMachine some ConstructionMachine` and `usesMaterial some Material`.

Featured Project – Trnava Smart Logistics & Assembly Hall (PRJ-IND-2025-01): Completed in `year = 2025` under Site Manager **Ing. Peter Horváth** for client *Central European AutoLogistics* (CUST-105). The hall spans 38,000 m² with a clear internal height of 14.5 meters. It utilizes prefabricated prestressed `Concrete` columns, 32-meter `Steel` roof trusses, PIR thermal `Insulation` sandwich panels, and a fiber-reinforced heavy-duty `Concrete` floor slab rated for 80 kN/m² point loads. Total budget: EUR 29,400,000.

6.2 Reconstruction Projects (Class: Reconstruction)

A critical architectural feature of `stavebna_firma_ontology - kópia.owl` is how `Reconstruction` is positioned in the class hierarchy:

> **Key Ontological Distinction (`Reconstruction` vs. `BuildingProject`):**
> Whereas `ResidentialProject`, `CommercialProject`, and `IndustrialProject` are subclasses of `BuildingProject`, the class **`Reconstruction` is a direct subclass of `Project`** (`Reconstruction ⊑ Project`), sibling to `BuildingProject`!
> `Reconstruction` explicitly requires:
>
> - `usesMachine some ConstructionMachine`
> - `usesMaterial some Material`
>
> Because it is a direct subclass of `Project`, it can represent not only building renovations, but also civil infrastructure rehabilitations (bridges, tunnels, retaining walls, and industrial brownfields).

Many of ApexBuild's reconstruction projects involve upgrading structures originally built during the **`MidCenturyProject`** period (defined by the ontology equivalence `1950 ≤ year < 2000`) or the **`HistoricalProject`** period (`year < 1950`):

Project Code	Project Name	Original Year & Period	Scope of `Reconstruction` Works
PRJ-REC-01	**Old Textile Mill Loft Conversion (Trenčín)**	Original: `1928` (`HistoricalProject`)	Underpinning historical `Brick` masonry vaults with micro-piles, inserting a secondary self-supporting `Steel` frame, and adding triple-glazed `Glass` roof skylights.
PRJ-REC-02	**Prievidza District Hospital Thermal Retrofit**	Original: `1974` (`MidCenturyProject`)	Asbestos abatement, structural carbon-fiber strengthening of `Concrete` panels, 200 mm mineral wool facade `Insulation`, and complete electrical rewiring.

6.3 Future Projects Pipeline: 2027 and Beyond (Class: FutureProject)

In the ontology, `FutureProject` is a subclass of `ProjectPeriod` defined via an `owl:equivalentClass` restriction as any project whose `year` property satisfies:

> **Formal OWL Axiom for `FutureProject`:**
> `FutureProject ≡ (year some integer[≥ 2027])` (`xsd:minInclusive 2027`).

ApexBuild Construction Corp. has already contracted three flagship projects scheduled for completion in 2027–2029, all classified under `FutureProject`:

1. **Košice GreenBattery Gigafactory Hall B (`year = 2027`):** A 65,000 m² clean-room industrial facility utilizing low-carbon geopolymer `Concrete` and recycled green `Steel` (Budget: EUR 94,000,000).
2. **Bratislava Port Eco-Tower (`year = 2028`):** A 32-storey hybrid timber-concrete commercial tower combining a `Concrete` core with structural cross-laminated `Wood` floors and photovoltaic `Glass` spandrels (Budget: EUR 68,000,000).
3. **Žilina Autonomous Rail Logistics Terminal (`year = 2029`):** Automated intermodal terminal featuring heavy-duty crane gantries and solar-heated warehouse slabs (Budget: EUR 41,500,000).